

DocuSense - DILAS DocuSense Architecture

for the following description,

DocuSense is a layered intelligent document analysis system designed to process and organize financial and utility-related documents under explicitly separated architectural and processing constraints. The system processes receipts, utility bills, invoices, and related document inputs obtained through on-device image capture, PDF import, or photo selection using a structured pipeline consisting of OCR extraction, token reconstruction, field identification, document-type classification, temporal normalization, trend analysis, anomaly detection, and structured visualization generation. Architectural responsibilities are separated into distinct processing layers responsible for document ingestion, OCR processing, semantic extraction, analytical computation, historical aggregation, visualization coordination, and persistence management in order to reduce cross-layer coupling and preserve controlled responsibility boundaries. DocuSense integrates multiple locally indexed structured datasets, including utility pricing datasets, historical billing records, and normalized temporal trend repositories supporting time-series comparison, anomaly association, contextual pricing analysis, and structured trend interpretation. OCR processing is performed using Apple Vision, while AI-related functionality is restricted to document classification, semantic field extraction, anomaly detection, trend interpretation, and structured analytical inference executed through on-device Core ML models. User authentication functionality is optionally supported through Firebase Authentication using limited authentication information consisting of email address and user identifier data. No images, extracted text, or document contents are transmitted to external AI services or external analytical systems, and all document processing and analytical operations are executed locally on-device. By separating document ingestion, semantic extraction, analytical processing, contextual dataset integration, and persistence responsibilities into explicitly bounded architectural layers, DocuSense functions as a

structured privacy-preserving intelligent document analysis and financial trend interpretation system.

Apply DILAS to the system description by identifying the architectural layers required for the system and defining the responsibilities of each layer.

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Applying DILAS, DocuSense can be organized into the following architectural layers:

Layer	Responsibilities
Presentation Layer	Provides user-facing interfaces for document capture, PDF import, photo selection, document browsing, trend dashboards, anomaly alerts, authentication screens, and structured report visualization.
Application Layer	Coordinates document-processing workflows, including ingestion orchestration, OCR pipeline execution, classification workflow control, semantic extraction coordination, analysis requests, and result delivery to the UI.
Domain Layer	Defines core document concepts and business rules, including receipts, utility bills, invoices, billing fields, temporal records, anomaly definitions, pricing concepts, trend entities, and validation rules.
AI Processing Layer	Performs on-device intelligent processing, including document-type classification, semantic field extraction, anomaly detection, trend interpretation, and structured analytical inference using Core ML.
Data Layer	Manages structured datasets and local records, including utility pricing datasets, historical billing records, normalized temporal trend repositories, extracted document metadata, and time-series comparison data.
Infrastructure Layer	Provides technical services such as Apple Vision OCR, local persistence, PDF/image handling, Core ML runtime integration, Firebase Authentication support, and device-level storage access.

Under DILAS, the architecture separates document ingestion, semantic extraction, analytical processing, dataset integration, visualization, authentication, and persistence into bounded layers so that DocuSense remains modular, locally executable, and privacy-preserving.

Applying DILAS, the architectural elements described in the system specification can be organized into the following components and assigned to the most appropriate layer.

Layer	Components
Presentation Layer	DocumentCaptureView, PDFImportView, PhotoSelectionView, DocumentBrowserView, TrendDashboardView, SpendingAnalysisView, AnomalyAlertView, AuthenticationView, ReportVisualizationView, ChartRenderer
Application Layer	DocumentIngestionController, OCRWorkflowCoordinator, DocumentProcessingOrchestrator, ClassificationCoordinator, SemanticExtractionCoordinator, TrendAnalysisCoordinator, AnomalyAnalysisCoordinator, AuthenticationCoordinator, ReportGenerationCoordinator
Domain Layer	Receipt, UtilityBill, Invoice, DocumentRecord, ExtractedField, BillingEntry, PricingRecord, TrendRecord, TimeSeriesModel, AnomalyRecord, DocumentType, TemporalNormalizer, PricingAnalysisRules, TrendAnalysisRules, AnomalyDetectionRules
AI Processing Layer	DocumentClassifier, SemanticFieldExtractor, AnomalyDetectionEngine, TrendInterpretationEngine, StructuredInferenceEngine, CoreMLInferenceManager
Data Layer	UtilityPricingRepository, HistoricalBillingRepository, TemporalTrendRepository, DocumentRepository, ExtractedDataRepository, AnomalyRepository, TrendRepository, PricingDatasetManager, HistoricalAggregationService
Infrastructure Layer	AppleVisionOCRService, OCRTextExtractor, TokenReconstructionService, LocalDatabaseManager, FileStorageManager, PDFProcessingService, ImageProcessingService, CoreMLRuntimeAdapter, FirebaseAuthenticationAdapter, SecureStorageService, DevicePersistenceManager

Component Responsibilities by Layer

Presentation Layer

Responsible for user interaction and visualization.

- **DocumentCaptureView** – capture documents using the device camera.
- **PDFImportView** – import PDF documents.
- **PhotoSelectionView** – select images from the photo library.

- **DocumentBrowserView** – browse processed documents.
- **TrendDashboardView** – display financial trends.
- **SpendingAnalysisView** – present spending summaries.
- **AnomalyAlertView** – display detected anomalies.
- **AuthenticationView** – login and account management.
- **ReportVisualizationView** – present analytical results.
- **ChartRenderer** – render graphs and trend visualizations.

Application Layer

Responsible for workflow orchestration and use-case coordination.

- **DocumentIngestionController**
- **OCRWorkflowCoordinator**
- **DocumentProcessingOrchestrator**
- **ClassificationCoordinator**
- **SemanticExtractionCoordinator**
- **TrendAnalysisCoordinator**
- **AnomalyAnalysisCoordinator**
- **AuthenticationCoordinator**
- **ReportGenerationCoordinator**

These components coordinate processing activities without implementing business rules directly.

Domain Layer

Responsible for business concepts and analytical rules.

Entities

- Receipt
- UtilityBill
- Invoice
- DocumentRecord
- ExtractedField
- BillingEntry
- PricingRecord
- TrendRecord
- TimeSeriesModel
- AnomalyRecord
- DocumentType

Domain Services

- TemporalNormalizer
- PricingAnalysisRules
- TrendAnalysisRules
- AnomalyDetectionRules

AI Processing Layer

Responsible for all AI-enabled analytical functionality.

- **DocumentClassifier**
- **SemanticFieldExtractor**
- **AnomalyDetectionEngine**
- **TrendInterpretationEngine**
- **StructuredInferenceEngine**
- **CoreMLInferenceManager**

All machine-learning inference is isolated within this layer.

Data Layer

Responsible for structured data management and aggregation.

- **UtilityPricingRepository**
- **HistoricalBillingRepository**
- **TemporalTrendRepository**
- **DocumentRepository**
- **ExtractedDataRepository**
- **AnomalyRepository**
- **TrendRepository**
- **PricingDatasetManager**
- **HistoricalAggregationService**

These components provide access to local datasets and historical records.

Infrastructure Layer

Responsible for platform and technology-specific implementation.

- **AppleVisionOCRService**
- **OCRTextExtractor**
- **TokenReconstructionService**
- **LocalDatabaseManager**
- **FileStorageManager**
- **PDFProcessingService**
- **ImageProcessingService**
- **CoreMLRuntimeAdapter**
- **FirebaseAuthenticationAdapter**
- **SecureStorageService**
- **DevicePersistenceManager**

These components encapsulate external frameworks, operating-system services, storage mechanisms, OCR technology, and authentication infrastructure.

Apply DILAS to define allowed and forbidden dependencies among the identified layers and components.

Using DILAS, dependencies are controlled through layer-based constraints that preserve separation of concerns and prevent architectural erosion. The resulting dependency structure for DocuSense is shown below.

Layer Dependency Rules

Source Layer	Allowed Dependencies	Forbidden Dependencies
Presentation	Application, Domain	AI Processing, Data, Infrastructure
Application	Domain, AI Processing, Data	Presentation, Infrastructure
Domain	None	Presentation, Application, AI Processing, Data, Infrastructure
AI Processing	Domain, Data	Presentation, Application, Infrastructure
Data	Domain	Presentation, Application, AI Processing, Infrastructure
Infrastructure	None	Presentation, Application, Domain, AI Processing, Data

Component-Level Dependencies

Presentation Layer

Component	Allowed Dependencies
DocumentCaptureView	DocumentIngestionController
PDFImportView	DocumentIngestionController
PhotoSelectionView	DocumentIngestionController
DocumentBrowserView	DocumentRepository, DocumentRecord
TrendDashboardView	TrendAnalysisCoordinator, TrendRecord

Component	Allowed Dependencies
SpendingAnalysisView	TrendAnalysisCoordinator, PricingRecord
AnomalyAlertView	AnomalyAnalysisCoordinator, AnomalyRecord
AuthenticationView	AuthenticationCoordinator
ReportVisualizationView	ReportGenerationCoordinator
ChartRenderer	TrendRecord, PricingRecord

Forbidden

- Direct access to repositories.
- Direct invocation of AI components.
- Direct access to OCR services.
- Direct access to persistence mechanisms.
- Direct access to Firebase or Apple Vision APIs.

Application Layer

Component	Allowed Dependencies
DocumentIngestionController	OCRWorkflowCoordinator, DocumentRecord
OCRWorkflowCoordinator	ExtractedField, DocumentRecord
DocumentProcessingOrchestrator	DocumentClassifier, SemanticFieldExtractor, DocumentRepository
ClassificationCoordinator	DocumentClassifier
SemanticExtractionCoordinator	SemanticFieldExtractor
TrendAnalysisCoordinator	TrendInterpretationEngine, HistoricalAggregationService, TrendAnalysisRules
AnomalyAnalysisCoordinator	AnomalyDetectionEngine, AnomalyDetectionRules
AuthenticationCoordinator	User domain entities
ReportGenerationCoordinator	TrendRecord, PricingRecord, AnomalyRecord

Forbidden

- Direct use of Apple Vision OCR APIs.
- Direct use of Firebase SDKs.

- Direct file-system access.
- Direct database access bypassing repositories.

Domain Layer

Component	Allowed Dependencies
Receipt	None
UtilityBill	None
Invoice	None
DocumentRecord	None
ExtractedField	None
BillingEntry	None
PricingRecord	None
TrendRecord	None
TimeSeriesModel	None
AnomalyRecord	None
DocumentType	None
TemporalNormalizer	Domain entities only
PricingAnalysisRules	Domain entities only
TrendAnalysisRules	Domain entities only
AnomalyDetectionRules	Domain entities only

Forbidden

- Repository access.
- OCR services.
- Core ML services.
- UI components.
- Framework-specific APIs.

The Domain layer remains independent and serves as the architectural center of the system.

AI Processing Layer

Component	Allowed Dependencies
DocumentClassifier	DocumentType, DocumentRepository
SemanticFieldExtractor	ExtractedField, DocumentType
AnomalyDetectionEngine	HistoricalBillingRepository, AnomalyRecord, TrendRecord
TrendInterpretationEngine	HistoricalBillingRepository, TemporalTrendRepository, TrendRecord
StructuredInferenceEngine	Domain entities, repositories
CoreMLInferenceManager	Domain models and repositories

Forbidden

- UI components.
- Controllers.
- OCR infrastructure services.
- Firebase services.

Data Layer

Component	Allowed Dependencies
UtilityPricingRepository	PricingRecord
HistoricalBillingRepository	BillingEntry, TrendRecord
TemporalTrendRepository	TrendRecord, TimeSeriesModel
DocumentRepository	DocumentRecord, ExtractedField
ExtractedDataRepository	ExtractedField
AnomalyRepository	AnomalyRecord
TrendRepository	TrendRecord
PricingDatasetManager	PricingRecord
HistoricalAggregationService	TrendRecord, BillingEntry

Forbidden

- Presentation components.
- Application controllers.
- AI inference engines.
- Infrastructure services.

Repositories depend only on domain entities.

Infrastructure Layer

Component	Allowed Dependencies
AppleVisionOCRService	None
OCRTextExtractor	None
TokenReconstructionService	None
LocalDatabaseManager	None
FileStorageManager	None
PDFProcessingService	None
ImageProcessingService	None
CoreMLRuntimeAdapter	None
FirebaseAuthenticationAdapter	None
SecureStorageService	None
DevicePersistenceManager	None

Forbidden

- Direct references to Presentation, Application, Domain, AI Processing, or Data components.

Infrastructure components act as technology adapters and remain isolated from business logic.

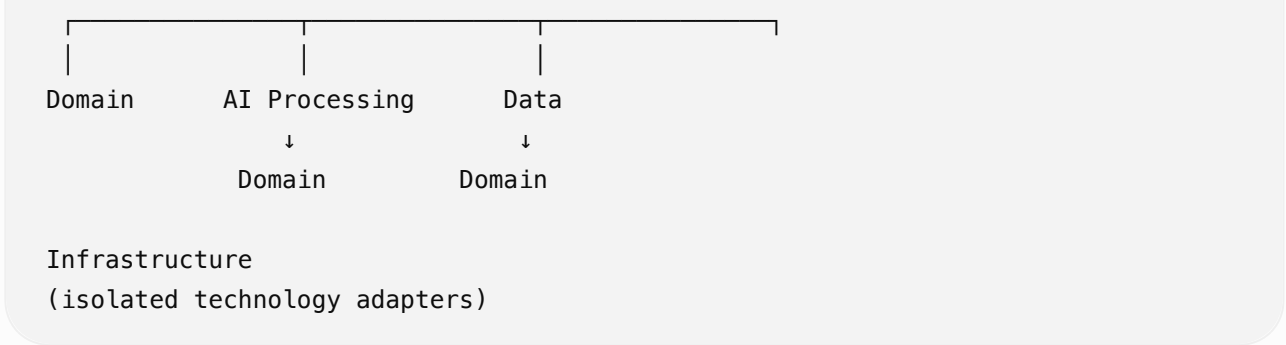
Resulting DILAS Dependency Structure

Presentation



Application





This dependency structure ensures that DocuSense maintains clear separation between user interaction, workflow orchestration, business rules, AI-based analysis, dataset management, and platform-specific implementation concerns while preserving the architectural constraints required by DILAS.



Apply DILAS to refine the internal structure of each layer by identifying major subcomponents and assigning responsibilities while preserving DILAS layer boundaries and dependency constraints.

Presentation Layer

Responsibility

Provides all user-facing interactions, document acquisition workflows, analytical result presentation, and visualization services.

Subcomponent	Responsibilities
Document Acquisition Module	Camera capture, photo selection, PDF import, document preview
Document Management Module	Document browsing, search, filtering, document history navigation
Analytics Dashboard Module	Spending summaries, billing trends, comparative analysis presentation
Anomaly Visualization Module	Anomaly alerts, anomaly explanations, anomaly history display
Reporting Module	Structured reports, trend summaries, analytical result presentation

Subcomponent	Responsibilities
Authentication UI Module	Login, logout, account management, authentication status display
Visualization Module	Charts, graphs, trend plots, comparative visualizations

Internal Components

Document Acquisition Module

- DocumentCaptureView
- PhotoSelectionView
- PDFImportView

Document Management Module

- DocumentBrowserView
- DocumentSearchView
- DocumentHistoryView

Analytics Dashboard Module

- TrendDashboardView
- SpendingAnalysisView
- ComparisonDashboardView

Anomaly Visualization Module

- AnomalyAlertView
- AnomalyDetailView

Reporting Module

- ReportVisualizationView
- SummaryReportView

Visualization Module

- ChartRenderer
- TrendChartRenderer
- ComparativeChartRenderer

Application Layer

Responsibility

Coordinates workflows, use cases, and interactions among domain, AI processing, and data services.

Subcomponent	Responsibilities
Document Processing Workflow Module	End-to-end document processing orchestration
Classification Workflow Module	Document classification coordination
Semantic Extraction Workflow Module	Field extraction coordination
Analytics Coordination Module	Trend and anomaly analysis orchestration
Authentication Coordination Module	Authentication workflow management
Reporting Coordination Module	Report generation and result preparation

Internal Components

Document Processing Workflow Module

- DocumentProcessingOrchestrator
- DocumentIngestionController
- OCRWorkflowCoordinator

Classification Workflow Module

- ClassificationCoordinator

Semantic Extraction Workflow Module

- SemanticExtractionCoordinator

Analytics Coordination Module

- TrendAnalysisCoordinator
- AnomalyAnalysisCoordinator
- HistoricalAnalysisCoordinator

Authentication Coordination Module

- AuthenticationCoordinator

Reporting Coordination Module

- ReportGenerationCoordinator
- DashboardCoordinator

Domain Layer

Responsibility

Defines business concepts, analytical abstractions, validation rules, and domain-specific computations.

Subcomponent	Responsibilities
Document Domain Model	Document entities and document semantics
Financial Domain Model	Billing, pricing, and spending abstractions
Temporal Analysis Domain	Time-series and trend abstractions
Anomaly Analysis Domain	Anomaly representation and anomaly rules
Domain Rules Module	Validation, normalization, and analytical rules

Internal Components

Document Domain Model

- DocumentRecord
- Receipt
- UtilityBill
- Invoice
- DocumentType
- ExtractedField

Financial Domain Model

- BillingEntry
- PricingRecord
- SpendingSummary

Temporal Analysis Domain

- TrendRecord
- TimeSeriesModel
- TrendSummary

Anomaly Analysis Domain

- AnomalyRecord
- AnomalyClassification

Domain Rules Module

- TemporalNormalizer
- PricingAnalysisRules
- TrendAnalysisRules
- AnomalyDetectionRules
- ValidationRules

AI Processing Layer

Responsibility

Performs all AI-enabled document understanding, analytical inference, anomaly detection, and trend interpretation.

Subcomponent	Responsibilities
Document Understanding Module	Classification and semantic extraction
Trend Intelligence Module	Trend interpretation and contextual analysis
Anomaly Intelligence Module	Anomaly detection and anomaly explanation
Inference Coordination Module	Shared inference execution and model management

Internal Components

Document Understanding Module

- DocumentClassifier
- SemanticFieldExtractor

Trend Intelligence Module

- TrendInterpretationEngine
- TrendInferenceEngine

Anomaly Intelligence Module

- AnomalyDetectionEngine
- AnomalyExplanationEngine

Inference Coordination Module

- StructuredInferenceEngine
- CoreMLInferenceManager
- ModelSelectionManager

Data Layer

Responsibility

Manages local datasets, repositories, historical information, and analytical data aggregation.

Subcomponent	Responsibilities
Document Data Module	Storage and retrieval of processed documents
Historical Billing Module	Historical billing and spending records
Trend Repository Module	Time-series and trend datasets

Subcomponent	Responsibilities
Pricing Dataset Module	Utility pricing and contextual pricing information
Analytics Data Module	Aggregation and analytical data preparation

Internal Components

Document Data Module

- DocumentRepository
- ExtractedDataRepository

Historical Billing Module

- HistoricalBillingRepository

Trend Repository Module

- TrendRepository
- TemporalTrendRepository

Pricing Dataset Module

- UtilityPricingRepository
- PricingDatasetManager

Analytics Data Module

- AnomalyRepository
- HistoricalAggregationService
- TrendAggregationService

Infrastructure Layer

Responsibility

Provides platform-specific services, framework integrations, storage implementations, OCR execution, AI runtime support, and authentication infrastructure.

Subcomponent	Responsibilities
OCR Infrastructure Module	OCR execution and token reconstruction
Document Processing Infrastructure Module	PDF, image, and file processing
Persistence Infrastructure Module	Database and storage implementation
AI Runtime Infrastructure Module	Core ML execution environment
Authentication Infrastructure Module	Firebase integration

Subcomponent	Responsibilities
Security Infrastructure Module	Secure storage and privacy protection
Internal Components	
OCR Infrastructure Module	
- AppleVisionOCRService - OCRTextExtractor - TokenReconstructionService	
Document Processing Infrastructure Module	
- PDFProcessingService - ImageProcessingService - FileStorageManager	
Persistence Infrastructure Module	
- LocalDatabaseManager - DevicePersistenceManager	
AI Runtime Infrastructure Module	
CoreMLRuntimeAdapter	
Authentication Infrastructure Module	
FirebaseAuthenticationAdapter	
Security Infrastructure Module	
- SecureStorageService - PrivacyProtectionManager	

Refined DILAS Structure

Presentation

- └ Document Acquisition
- └ Document Management
- └ Analytics Dashboard
- └ Anomaly Visualization
- └ Reporting
- └ Authentication UI

Application

- └ Document Processing Workflow
- └ Classification Workflow
- └ Semantic Extraction Workflow
- └ Analytics Coordination
- └ Reporting Coordination
- └ Authentication Coordination



Domain

- └ Document Domain Model
- └ Financial Domain Model
- └ Temporal Analysis Domain
- └ Anomaly Analysis Domain
- └ Domain Rules

AI Processing

- └ Document Understanding
- └ Trend Intelligence
- └ Anomaly Intelligence
- └ Inference Coordination

Data

- └ Document Data
- └ Historical Billing
- └ Trend Repository
- └ Pricing Dataset
- └ Analytics Data

Infrastructure

- └ OCR Infrastructure
- └ Document Processing Infrastructure
- └ Persistence Infrastructure
- └ AI Runtime Infrastructure
- └ Authentication Infrastructure
- └ Security Infrastructure

This refinement preserves DILAS boundaries by ensuring that presentation concerns remain in the Presentation layer, workflow orchestration remains in Application, business semantics remain in Domain, machine-learning functionality remains in AI Processing, dataset management remains in Data, and all framework- and platform-specific implementations remain isolated within Infrastructure.

